Life Sciences and
Facility Management

Institute of
Computational Life Sciences

Process variables & Parameter, KPIs, BPMN

Simulation for Beginners

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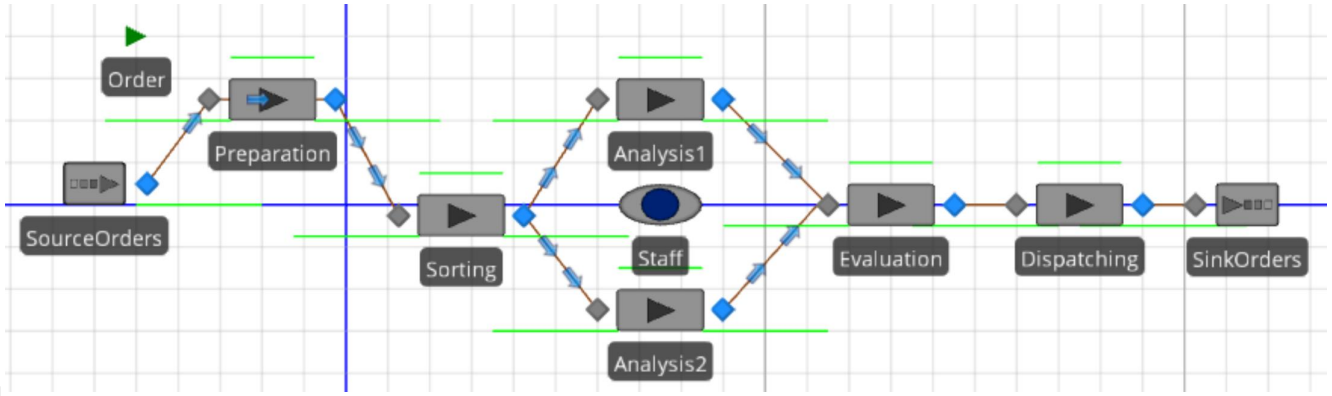
Content



- **Model Elements**
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- **Process Variables and Parameters**
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- **Key Performance Indicators (KPIs)**
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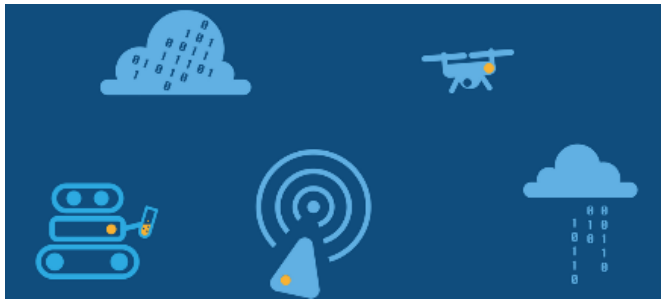
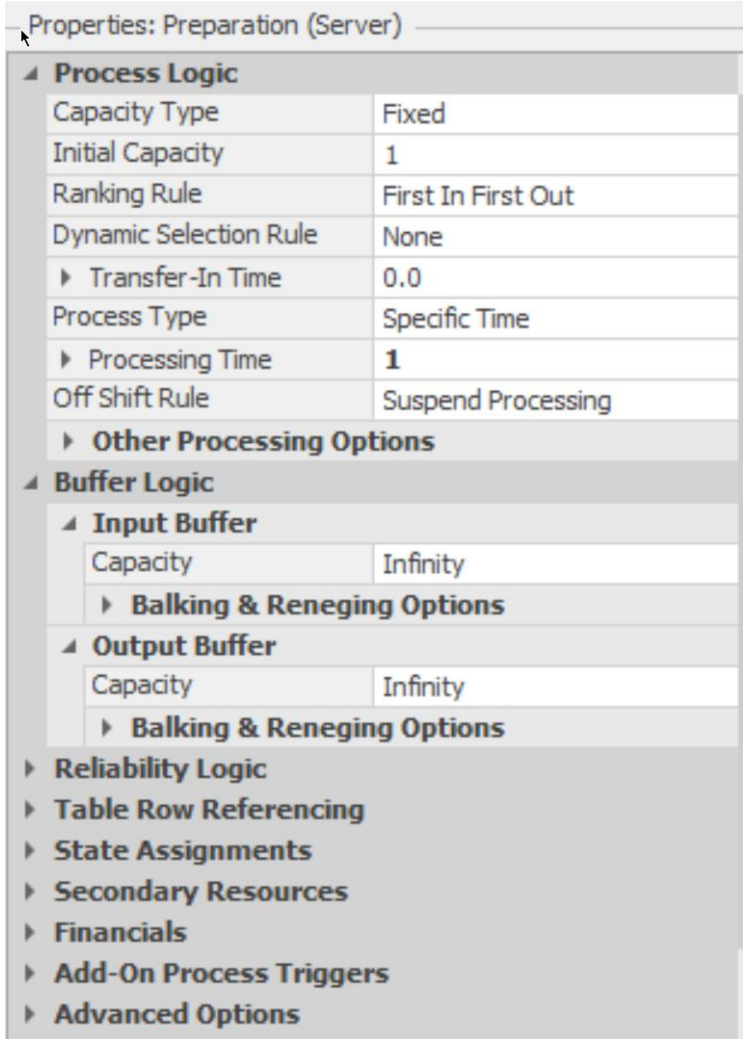
- **Business Model Process Notation (BPMN)**



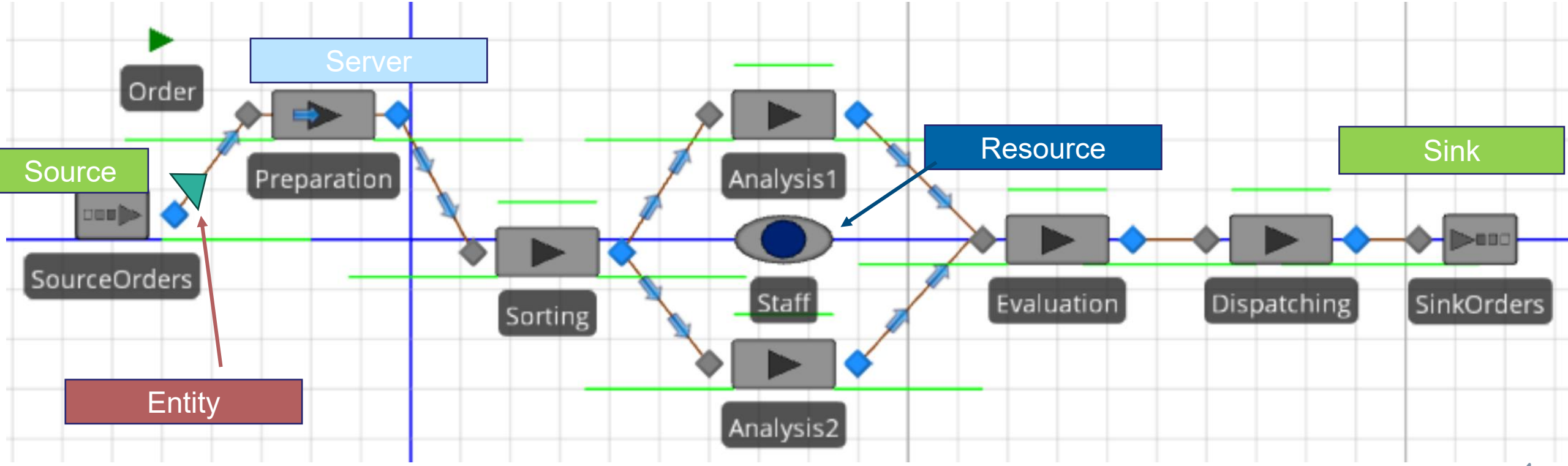
Modeling concepts

Model elements

- **Moving Elements (Entities)**
  Flow through the model. They are processed and transported, physically or virtually.
- **Source / Sink**
  In sources, moving elements are created. In sinks, the moving elements are "destroyed."
- **Server (Active Service Station)**
  Non-moving model elements that provide capacity. Moving elements stay here.
- **Resources**
  Static or dynamic. Required for processing or transporting moving elements and also provide capacity.
- **Queues**
  Buffer areas where units (e.g., customers, products, or orders) wait for processing by limited resources, allowing bottlenecks and system utilization to be analyzed.



Model Elements Example



Example: Laboratory Simulation

Specification / Data

Process steps

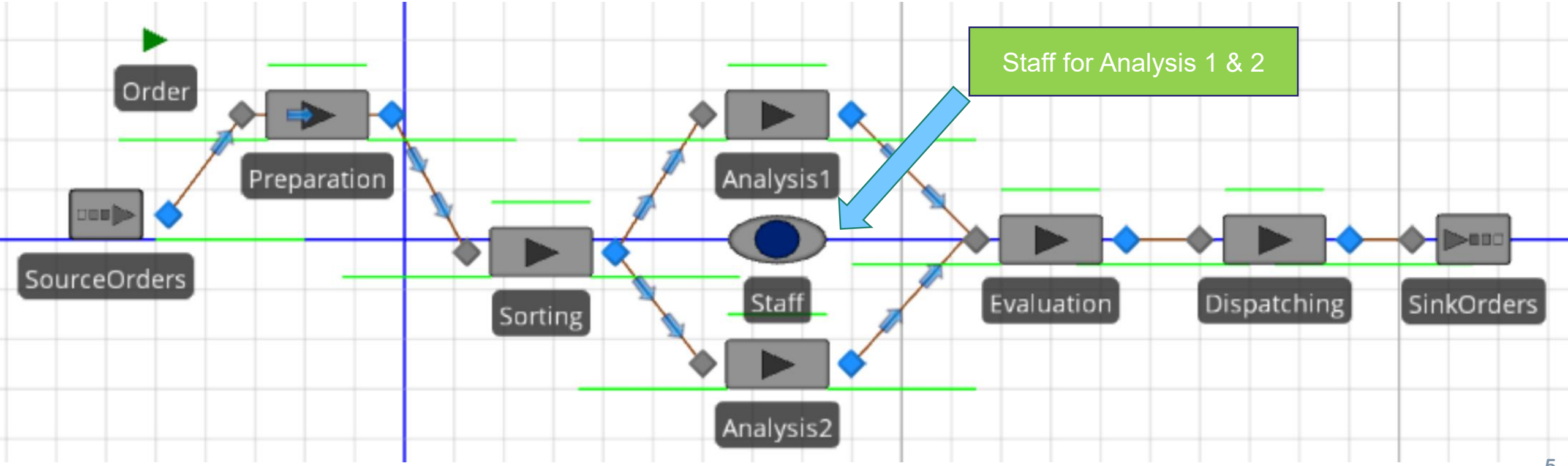
Preparation
- 1.5 – 2.5 min
- 1 Staff

Sorting
- 3-5 min
- 1 Device

Analysis
- Analysis 1 (80%)
  - 3 – 6 min, 1 Device
- Analysis 2 (20%)
  - 2 min, 1 Device

Evaluation
- 0.2 – 0.75 min
- 1 Device

Dispatching
- 0.8 – 1.2 min.
- 1 Mitarbeiter

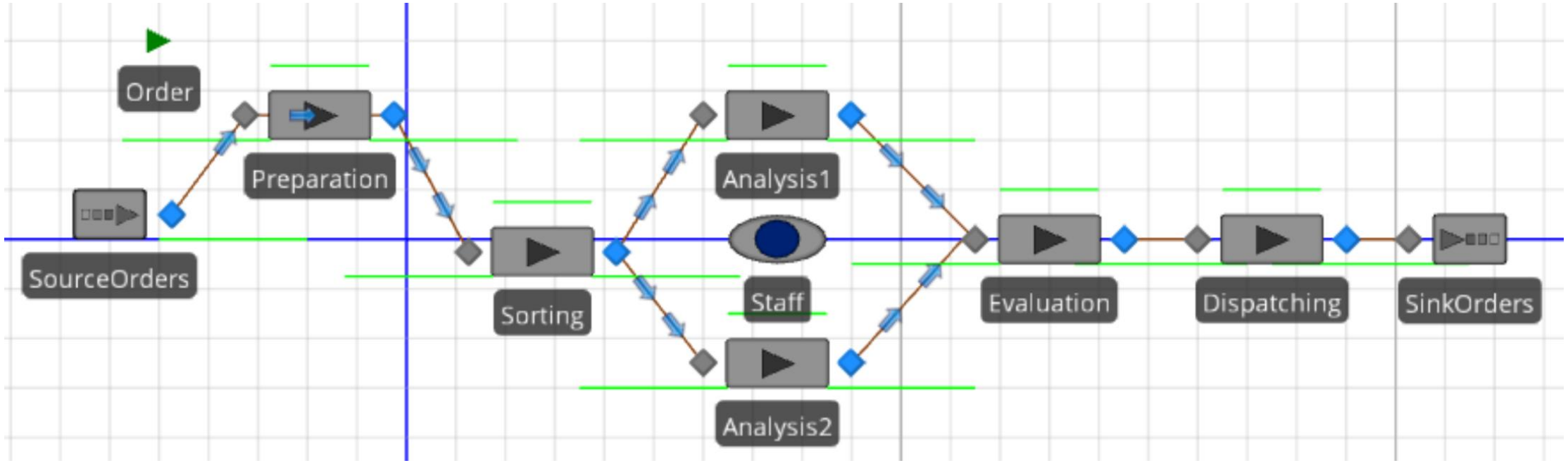


Common process variables for various element types

- Moving Elements (Entities)
- Source / Sink
- Server (Active Service Station)
- Resources
- Queues

Control Concepts

- **Decision Points:**
  - **Branches:** How is the distribution done?
  - **Merging:** Prioritization?
  - **Shared Resource:** Prioritization?



Random variables: Triangular Distribution

- **Simple Parameter Structure:**
  - Defined by three parameters – **Minimum, Mode** (the most likely value), and **Maximum** – making it particularly easy to specify.
- **Modeling Uncertainties:**
  - In process simulation, it is well-suited for representing uncertainties and variability, especially when only limited historical data is available.
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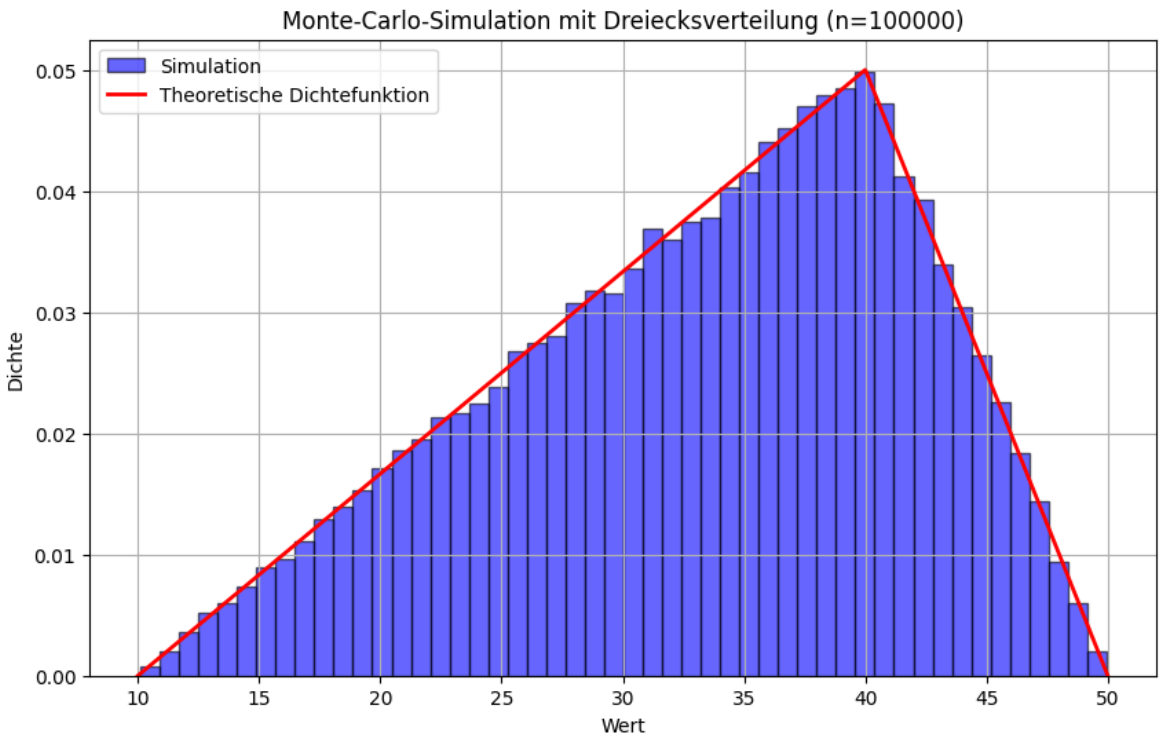
- **Realistic Representation:**
  - The distribution emphasizes the most likely value, which often enables a more realistic depiction of expected values in simulation models.



Experiments / Results / Evaluations

Prozessvariablen und Parameter

Parameter	Type	Unit	Key	
Scenario number	int		scenario	1
Scenario name	str		scenario_name	1
Run duration	float	min	run_duration	1440
Use hourly arrival rates	bool		use_hourly_arrival_rates	WAHR
Rate multiplier	float	1	rate_multiplier	1
Analysis1 fraction	float	1	analysis1_fraction	0.6
Analysis2 after 1 fraction	float	1	analysis2_post1_fraction	0.3
Worker capacity	int	1	worker_capacity	1
Preparation capacity	int	1	preparation_capacity	1
Sorting capacity	int	1	sorting_capacity	1
Analysis1 capacity	int	1	analysis1_capacity	4
Analysis2 capacity	int	1	analysis2_capacity	4
Evaluation capacity	int	1	evaluation_capacity	1
Dispatching capacity	int	1	dispatching_capacity	1
Preparation process duration Low	float	min	preparation_pt_low	1.5
Preparation process duration High	float	min	preparation_pt_high	2.5
Preparation process duration Mode	float	min	preparation_pt_mode	2
Sorting process duration Low	float	min	sorting_pt_low	3
Sorting process duration High	float	min	sorting_pt_high	5
Sorting process duration Mode	float	min	sorting_pt_mode	4

Arrival rates

Random variables

Capacities

Process duration

Experiments / Results / Evaluations

KPIs

Objectives

- Proof of functionality of the planned system
- Determining the performance limits of the system
- Identifying bottlenecks and developing measures to eliminate them
- Demonstrating how the system behaves under peak load
- Determining the utilization levels of individual system components
- Assessing the impact of different workload scenarios
- Determining the influence of individual system parameters on overall performance
- Evaluating the impact of disturbance scenarios on system performance (e.g., failure of an analysis device, etc.)

KPIs

- Throughput (at various measurement points and at the sinks)
- Utilization of individual system components (equipment, employees)
- Lead times
- Inventory levels
- Stock balance

Business Process Model Notation (BPMN)

- Standard for graphical process modeling
- Flowchart with standardized symbols
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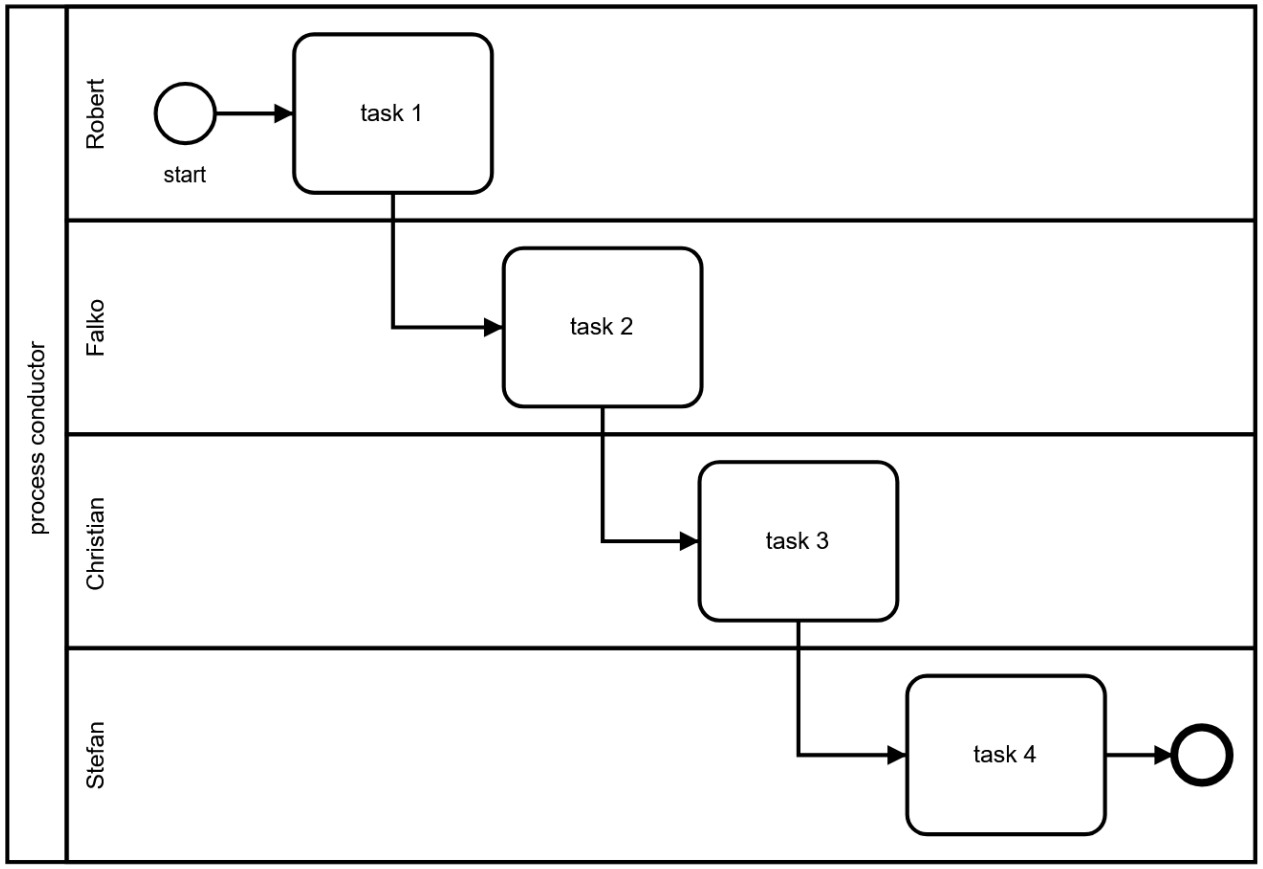
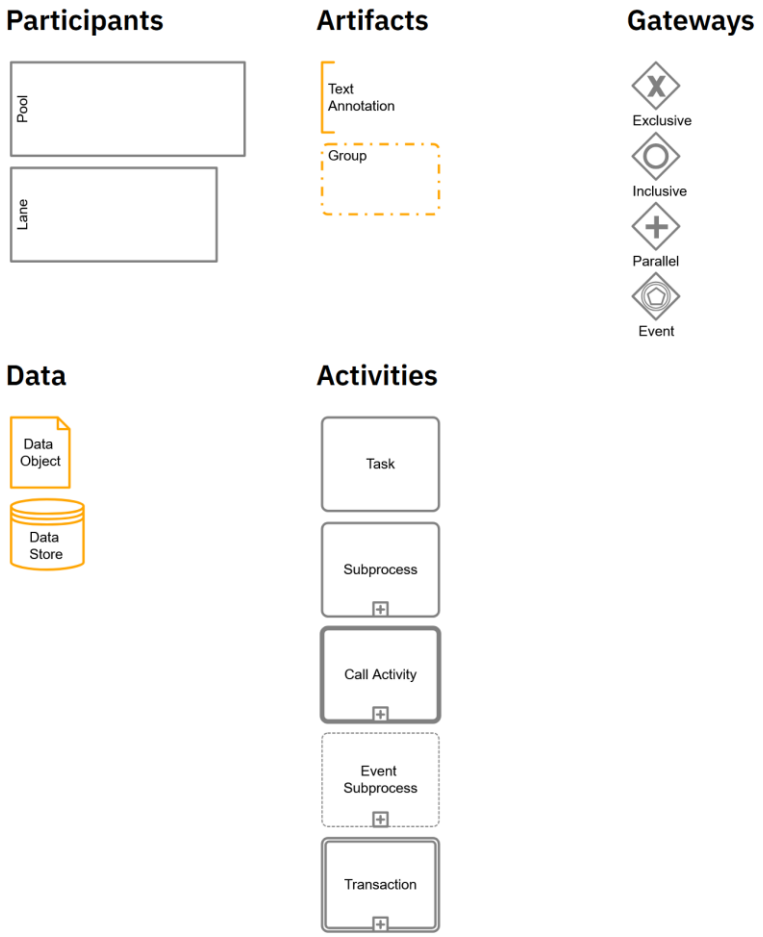
- **Comprehensibility** → clear, uniform presentation of business processes
- **Simulation suitability** → processes can be tested before implementation
- **Extensibility** → addition of simulation parameters (e.g. times, resources)

Introduction to BPMN:
https://www.sciencedirect.com/science/article/pii/S0920548911000766

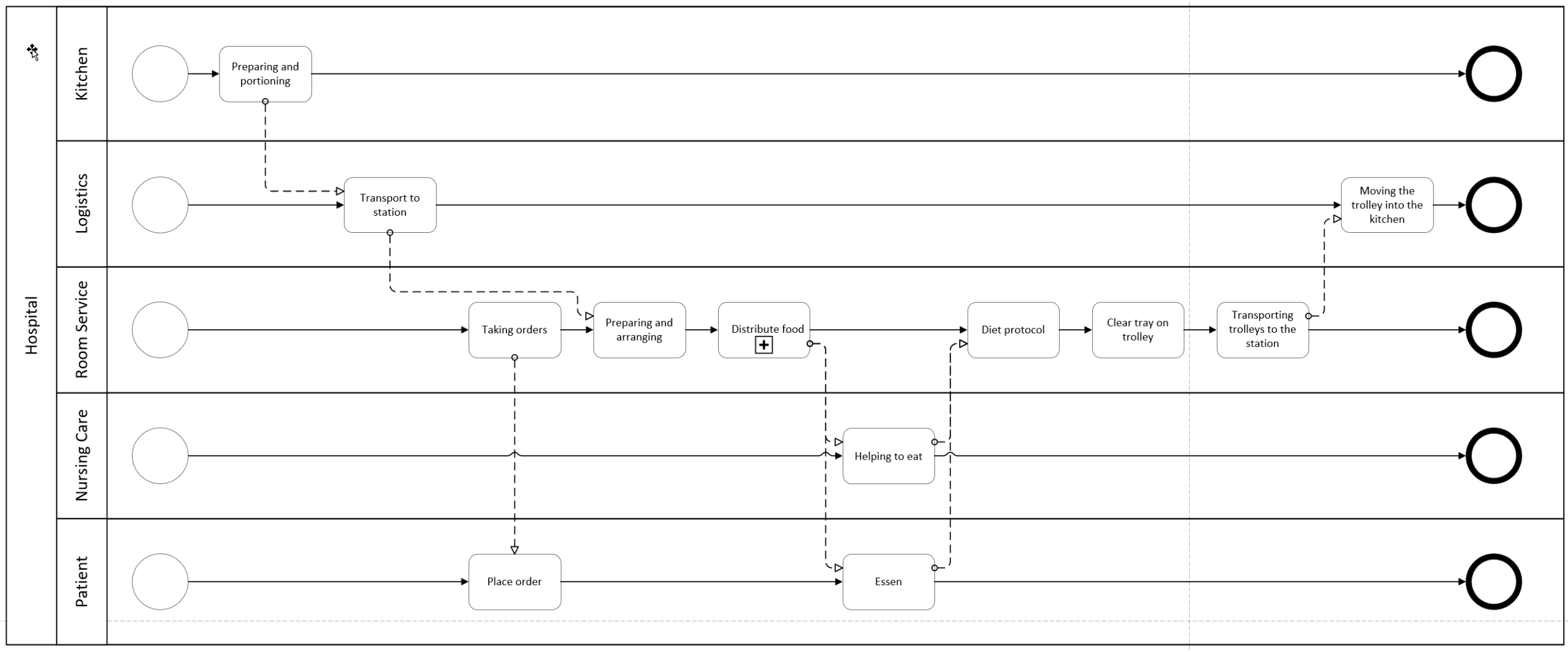
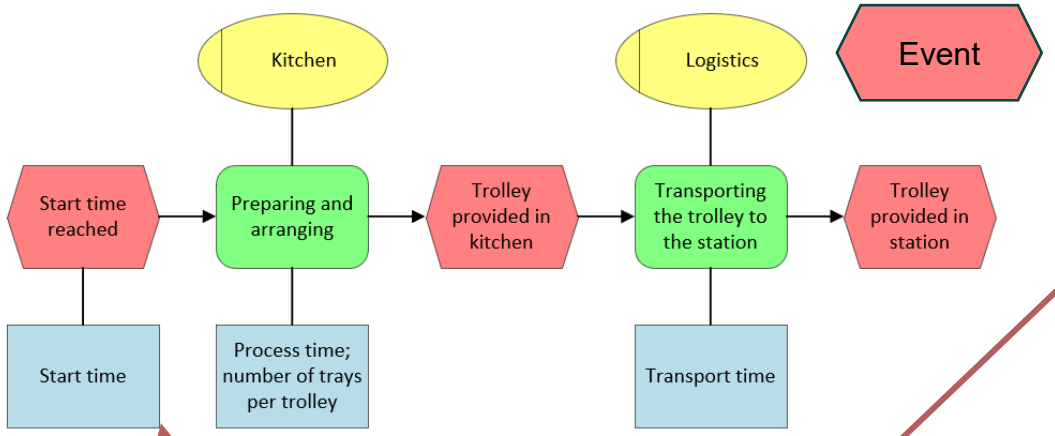
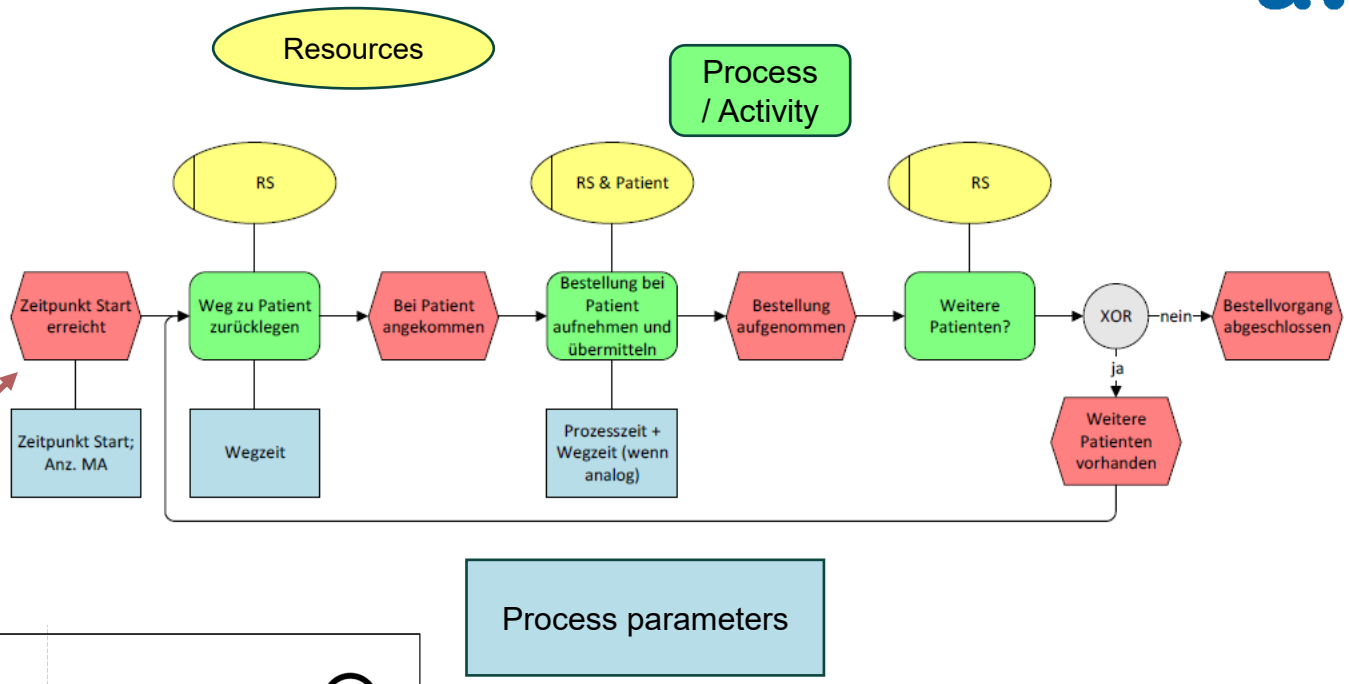
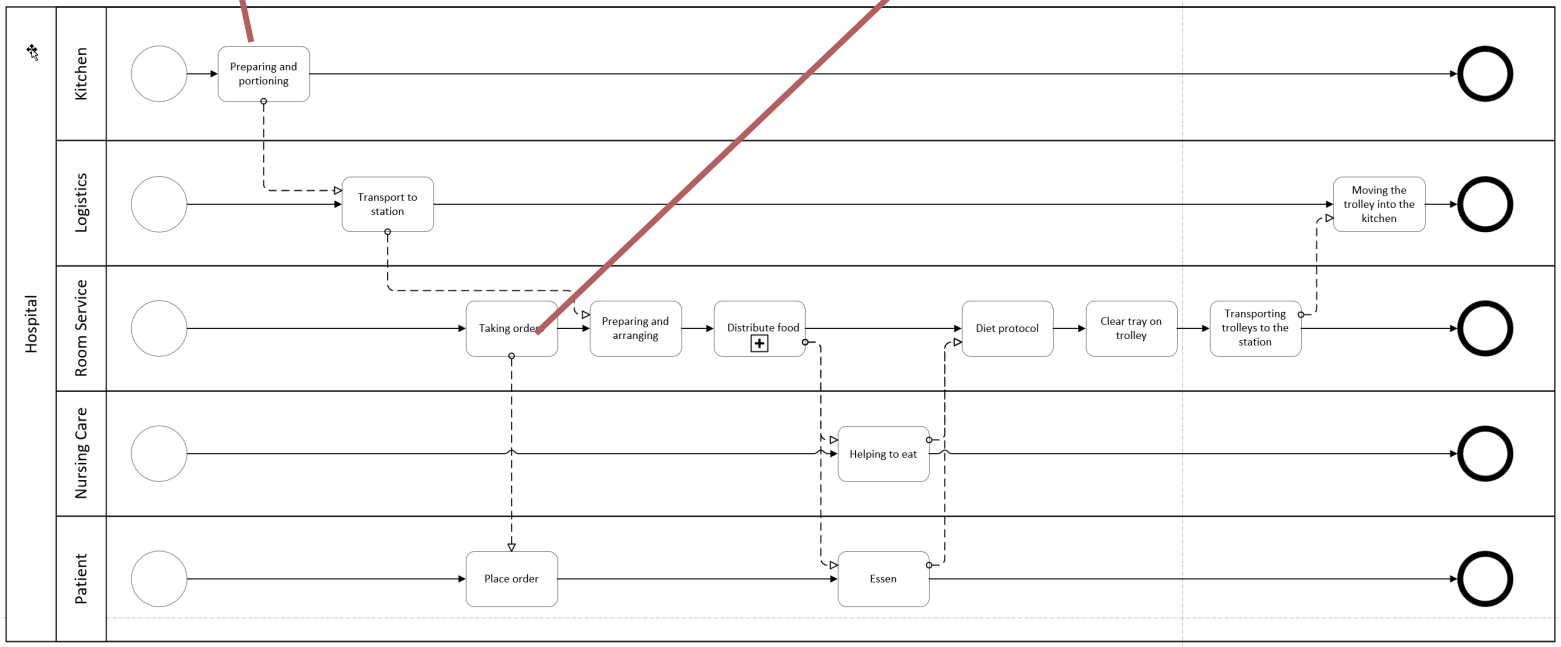
Reference symbols





Quelle: https://camunda.com/de/bpmn/bpmn-2-0-symbol-reference/

Catering in hospital



Catering in the hospital